LethalScanner

WEB SECURITY SCANNER

SYSTEM ANALYSIS AND DESIGN

PROJECT PROPOSAL PRESENTATION BY ABINASH REGMI

ABSTRACT

In today's digital world, web security is very important because many web applications unintentionally expose sensitive directories and configuration files. These hidden paths can lead to Information Disclosure where unauthorized users access private data. LethalScanner uses Python, Requests, and SQLite. It analyzes HTTP status codes like 200 OK and 403 Forbidden. Findings are stored in a structured database. It aautomates web reconnaissance and improves security.

Keywords: *Web Security, Information Disclosure, Python, Requests Library, SQLite, Automation, Brute-forcing.*

PROBLEM STATEMENT

Current security tools often present a two choice: expensive enterprise ventures or limited command-line scripts. There is a lack of accessible tools that combine high speed multi-threaded scanning with structured data persistence. Without persistence, critical discoveries made during an audit are often lost, requiring auditors to repeat tasks and wasting valuable time. In short:

1. **Lack of accessible security tools**
2. **Slow manual scanning**
3. **No structured reporting**
4. **Expensive enterprise solutions**

OBJECTIVES

The main objectives of the LethalScanner System are as follows:

To automate the discovery of hidden directories and configuration files in web applications using Python libraries.

To detect information disclosure vulnerabilities efficiently by analysing HTTP status codes (e.g., 200 OK, 403 Forbidden) and confirming the existence of sensitive paths.

To store and organize vulnerability findings in a structured SQLite database, enabling developers to generate clear reports of the web architecture.

INTRODUCTION

The **digital transformation** of global commerce and communication has let to an era where web applications are the primary interface for data exchange. However, this shift has also expanded the attack surface for **malicious entities**. Information has become a **valuable** thing. **Information Disclosure**, a vulnerability, where sensitive data is unintentionally **leaked** through server misconfigurations, has become a wide threat. **Lethal Scanner** is designed to address this specific problem of **cybersecurity** by providing a structured, **automated approach to web security**.

LITERATURE REVIEW

Web security tools such as Dirbuster and Gobuster are widely used to brute-force directories in web applications. These tools are effective but rely on programming languages like Java and Go, which can limit portability and ease of use for beginners [1].

Other scanners, such as Nikto, are designed to detect outdated software versions and misconfigurations. While Nikto is useful for identifying common vulnerabilities, it does not provide persistent reporting or structured data storage, which reduces its long-term usability [2].

Wfuzz offers flexibility in brute-forcing parameters and directories, making it a powerful tool for penetration testers. However, its complexity requires advanced knowledge to configure, and its lack of built-in reporting features means users must manually manage results [3].

METHODOLOGY

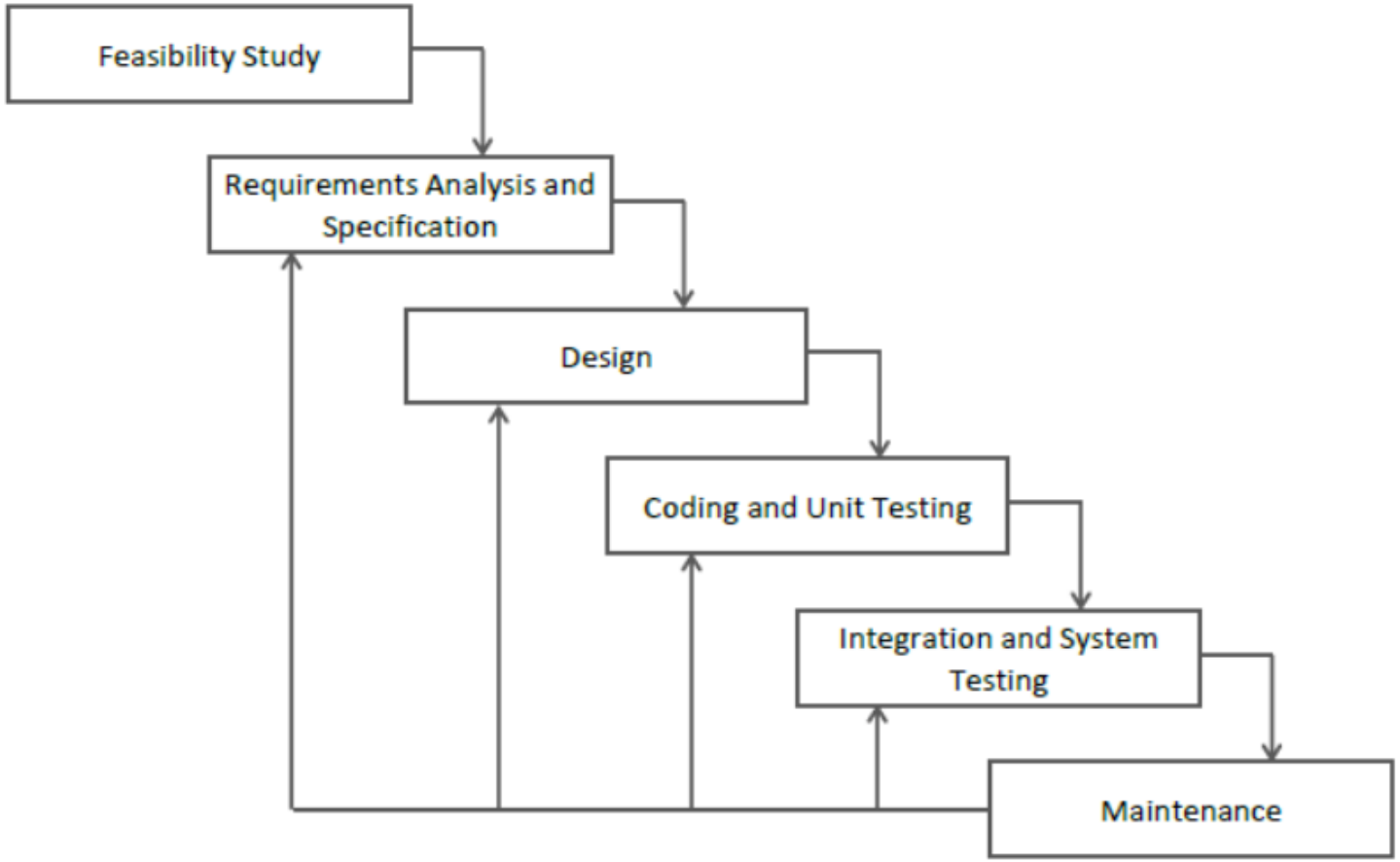


Figure 1: Iterative Waterfall Model

SYSTEM MODULES

🌐 **Target Input**

⚙ **Scanning Engine**

📊 **Result Analyzer**

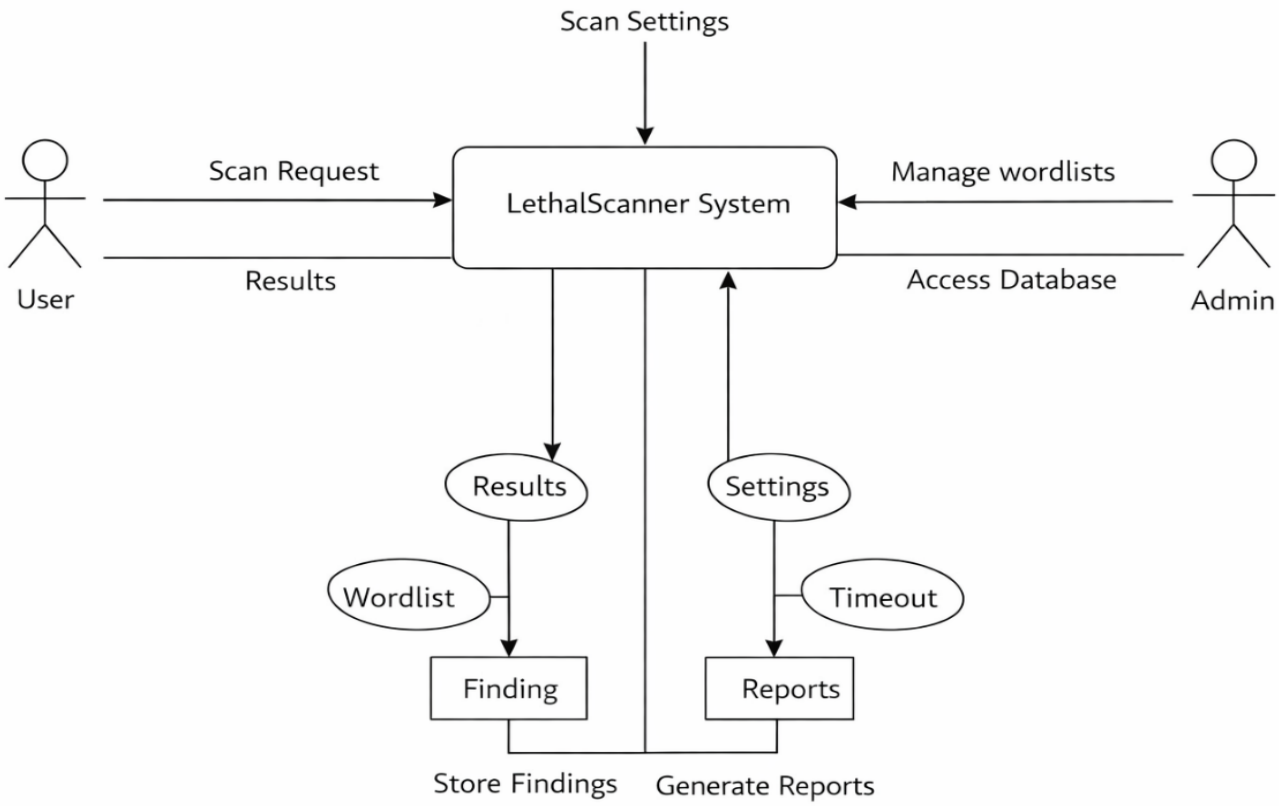
💾 **Database Storage**

📄 **Report Generator**

SYSTEM OVERVIEW

1. User inputs target URL
2. Scanner sends requests
3. Analyzes responses
4. Stores results in database
5. Generates report

CONTEXT DIAGRAM (level 0 DFD)



GANTT CHART

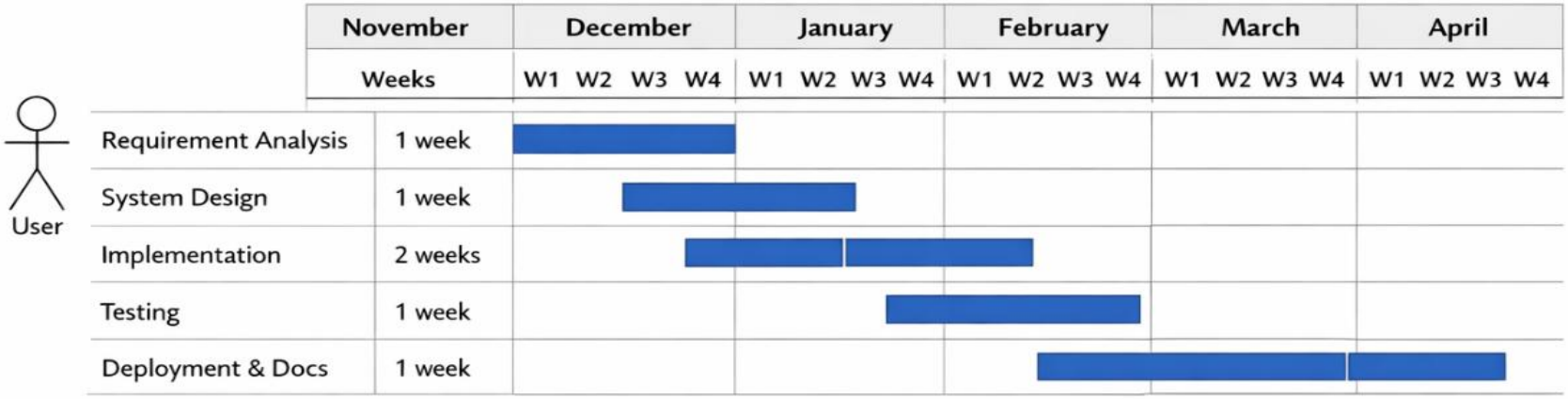


Table 1-1: Gantt Chart of Project Activities

EXPECTED OUTCOME

The main outcome of this project is a working tool called LethalScanner that can scan websites to find hidden folders and files. It will help users detect paths that may cause information leaks. The tool will show results clearly, save them in a database, and allow users to generate reports. It will be easy to use and helpful for basic security testing.

- ✅ **Functional tool**
- 🔐 **Detect vulnerabilities**
- ⚡ **Fast scanning**
- 📊 **Structured reports**
- 📈 **Better security**

FUTURE RECOMMENDATIONS

01 Multi-threading

02 GUI Interface

03 Multi-target scanning

04 Advanced detection

05 Cloud deployment

CONCLUSION

1. Efficient web security scanner
2. Automates vulnerability detection
3. Helps prevent attacks early

REFERENCES

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Thank You!

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Any Questions?